

STEPHEN MBUGUA

Machine Learning (MLOPS)Engineer| Data Science Enthusiast

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Professional Summary

Strategic MLOps Engineer finalizing an MBA, with 4+ years of applied data experience orchestrating end-to-end machine learning lifecycles. Expert in designing and implementing scalable MLOps pipelines on cloud platforms (AWS and OCI) to automate the deployment, monitoring, and governance of production-grade models. Proven ability to bridge the gap between data science and engineering, leveraging a unique blend of deep technical expertise including Generative AI, Lang Chain, Docker and Kubernetes and business acumen honed through immersive experiences at JP Morgan, Electronic Arts, and Commonwealth Bank. Driven to transform cutting-edge AI potential into reliable, high-impact business solutions.

Professional Development & Certification

Technical Certifications

- Oracle Cloud Infrastructure Foundations Associate | Oracle University
- Python Certificate - SQL, TensorFlow/PyTorch, Scikit-learn, Machine Learning, Deep Learning | Bright Networks Academy

Specialized Training

- MLOps & DevOps: Docker, Kubernetes, CI/CD, Git, ML Pipeline Design
- Data Science & Analytics Intensive Program | LuxTech Academy, Nairobi
(A comprehensive curriculum covering Excel, MySQL, Python, Machine Learning, and Deep Learning)

Finance & Markets

- International Introduction to Securities & Investments | Chartered Institute for Securities & Investment (CISI)

Experience

Machine Learning Engineer (Remote) | Data Science East Africa | January 2021 – Present

Orchestrated the end-to-end machine learning lifecycle, from initial data analysis and model development to the deployment and monitoring of production-scale MLOps pipelines. Leverages a strong foundation in data science and analytics to build robust, data-driven solutions for diverse Data Science East Africa projects.

- Engineered and automated continuous training and deployment (CI/CD) pipelines for machine learning models, reducing manual deployment time by an estimated **30%** and standardizing the release process across **5+ projects**.
- Designed and implemented sophisticated **AI agents** using **LangChain** and **LangGraph** to create complex, stateful workflows for reasoning, data analysis, and task automation, enhancing the capabilities of Data Science East Africa applications.
- Containerized machine learning models and dependencies using **Docker** and orchestrated deployments with **Kubernetes**, ensuring consistent environments from development to production and improving system scalability.
- Collaborated closely with data scientists to refactor experimental code into production-ready, efficient software, facilitating the seamless transition of **10+ models** from research to live environments.
- Implemented robust **model monitoring** and **governance frameworks** to track **performance metrics, data drift, and model accuracy in production**, enabling proactive maintenance and ensuring **long-term reliability**.
- Championed **MLOps best practices** within the team, including **version control for data and models** (via **Data Version Control** and **Git**), establishing a reproducible model lineage, and documenting workflows.

Lead Customer Experience Associate – Hotpoint Appliances Limited

Led strategic sales initiatives that expanded market reach, strengthened customer loyalty, and boosted monthly revenue. Identified high-value leads, developed client-specific proposals, and collaborated with internal teams to deliver optimal solutions that aligned with customer needs and business objectives. Responsible for driving revenue growth by managing key customer relationships, identifying new business opportunities, and delivering tailored product solutions. Oversaw the full sales cycle, from prospecting and negotiations to closing deals, while ensuring exceptional customer experience and consistent achievement of sales targets.

Nairobi | July / 2020– Present

- Help in daily store operations, ensuring smooth workflow and adherence to company policies.
- Develop and execute sales strategies to meet and exceed revenue targets.
- Analyze sales data to identify trends, opportunities, and areas for improvement.
- Recruit, train, schedule, and manage staff to maintain optimal performance levels.
- Lead and motivate teams through coaching, performance reviews, and sales incentives.
- Ensure excellent customer experience by resolving complex issues and maintaining a welcoming environment.
- Manage inventory levels, stock rotation, merchandising, and store presentation.
- Ensure store security, compliance, and brand standards are consistently upheld.
- Prepare store reports, sales forecasts, and operational updates for senior management.

Personal Projects

Generative AI & LLM Applications

- **Alzheimer's Patient Support AI (Hackathon Project)** | Python, TensorFlow, Speech-to-Text, LangChain
 - Collaborated in a time-constrained hackathon to design and prototype an AI-powered assistant that helps patients with Alzheimer's by checking their cognitive state, discerning confusion in conversation, and providing contextual reminders and support to improve daily living.
- **Constitutional RAG System** | Python, LangChain, LLMs
 - Developed a Retrieval-Augmented Generation (RAG) system based on the Constitution of Kenya, enabling efficient, accurate Q&A on complex legal documents to improve information accessibility.
- **Multimodal Query Agent** | Python, LangChain, AI Agents
 - Engineered an advanced AI agent capable of processing images, text documents, and URLs as input. The system synthesizes information from these multimodal sources to provide comprehensive answers to user queries.
- **Customer Service Chatbot** | Python, NLP, OpenAI API
 - Designed and implemented an automated chatbot to handle customer service inquiries, reducing response time and streamlining support workflows.

Corporate Immersion Simulations

- **Global Finance & Gaming Industry Simulations**
 - Completed intensive virtual experience programs with **Commonwealth Bank of Australia**, **JP Morgan**, and **Electronic Arts**.
 - Gained practical insight into real-world business challenges, including **financial analysis**, **strategic investment modeling**, and **digital marketing campaign** execution within regulated and fast-paced industries.

Skills

Programming & ML Frameworks

- **Languages:** Python (Certified, Bright Networks Academy), SQL
- **ML/Deep Learning:** Scikit-learn, TensorFlow, PyTorch
- **Generative AI & LLMs:** LangChain, LangGraph, AI Agents, RAG, OpenAI API
- **NLP:** Transformer Models, Speech-to-Text, Text Processing

MLOps & Cloud Platforms

- **MLOps Tools:** MLflow, DVC (Data Version Control), Kubeflow
- **CI/CD & Automation:** Jenkins, GitHub Actions, Automation Scripting
- **Containerization & Orchestration:** Docker, Kubernetes
- **Cloud Platforms:** Oracle Cloud Infrastructure (OCI Foundations Associate), AWS, Azure, GCP

Data Management & Engineering

- **Databases:** MySQL, Oracle DB
- **Data Analysis & Tools:** Pandas, NumPy, Excel
- **Data Pipelines:** ETL/ELT, Data Warehousing

Soft Skills & Business Acumen

- **Cross-Functional Collaboration:** Working with Data Scientists, Engineers, and Business Stakeholders
- **Strategic Thinking:** MBA-level business strategy, financial analysis, market modeling
- **Problem-Solving:** Complex issue troubleshooting, analytical reasoning
- **Project Management:** Agile Methodologies, Project Lifecycle Management

Industry Knowledge

- **Finance:** Securities & Investments (CISI Certified), Financial Modeling, Risk Management
- **Healthcare:** AI in Patient Support Systems, Medical Data Processing
- **Gaming & Entertainment:** Digital Marketing, User Engagement Analytics

Interests

- **Technology & Innovation:** Exploring open-source AI projects, attending tech meetups, and building personal automation scripts.
- **Strategic Games:** Playing complex strategy video games that challenge problem solving and long term planning.
- **Continuous Learning:** Reading non-fiction on behavioral economics and the history of technology.
- **Outdoors:** Hiking and landscape photography.

